Spring 2024 – User Research Methods – Professor Charles Raffaele
Video Observation Design Document REFLECTION – Darryl Hutchinson

REFLECTION INTRODUCTION

Overall, I think my video observation design helped get closer to answering the research questions I proposed to better understand what game elements effectively communicate to players. Having open-ended research questions with multiple potential interpretations highlights the value of using video observation with a specific coding scheme. Being able to access and manipulate the video data of participant behavior as well as having multi-dimensional scheme of codes, each of which are directly connected to an aspect of a research question, to more specifically analyze each event in the video data provides a stronger foundation to answering the research questions and, in turn, determining if and how well the game elements of Europa, or potentially other games using a similar research approach, communicates to the player. After conducting the video observation session and this reflection, I the strengths of video observation were more clear, but I also left room for improvement.

Just before the session started, I noticed a few things that may have benefit from adjustment. First of all, I didn't plan to emphasize think-aloud nor any warm-up activity. However, just before starting the session I felt that having the additional layer of verbal participant feedback could be helpful for the research question so I made a last minute adjustment, setup a warm up activity and requested that the participant think-aloud during the video observation. Similarly, I was so focused on all the setup for the video observation that I'd forgotten about setting up a survey/questionnaire for the participant. I once again felt that information could be helpful for these research questions, so although it was not a part of my original design document, I did decide to administer a pre-session survey and post-session questionnaire.

Also, I wasn't entirely sure about how best to organize the coding section. I already had a sample coding sheet in my original design document rom before we did the extra review in class. But, I still didn't feel confident that it was organized in a way the professors expects. The datavyu example was not so easy for me to parse and though it doesn't seem to be the intended format of this course, I might have benefit from getting feedback on the original video observation design document before submitting this reflection. Still, I did try add more detail and examples to this reflection version of the sample coding sheet. Either way, with such confusion still in mind, I should have asked more questions.

A. INTRODUCTION

Though they have great potential for entertainment and learning, video games are typically very complex and it can be easy for inexperienced players to get overwhelmed and lose site of the objectives the designer wants them to experience. To better understand intuitive level design and which game elements do the best job of capturing player attention regardless of experience level such that they don't get lost and can progress through an experience, I want to analyze the upcoming video game, Europa.

Designed and developed by Helder Pinto and Novadust studios, Europa is a bright and colorful Sci-Fi adventure game with a heavy influence of painterly Ghibli style visual animation, smooth freeform

movement and traversal with low stress exploration. The game has been in development since 2017 and is currently available in demo form with a scheduled release window of spring 2024.

I am interested in exploring pacing/pathing and environmental storytelling. I would like to know specifically how game elements in the game world can communicate which ways players can travel without using text or unnatural UI markers. Since video games are relatively complex bundles of systems that are easy to get lost within, I want to try and understand how to better lead the player along an experience with natural elements that help maintain immersion.

The target audience for this research could be the developers Novadust studios themselves to gain more perspective on what their players are experiencing. With more feedback about what is drawing player attention, adjustments can be made to the level design to further avoid players getting lost or confused in the game. Another audience could also be other developers, researchers and/or game design enthusiasts who want to more deeply understand how specific elements in Europa create specific responses or perceptions in players.

This analysis could easily be applied to another game, but I am choosing the same game as the previous think-aloud protocol analysis in order to keep as many variables as consistent as possible to more clearly compare/contrast the differences in observation approach. Highlighting these differences will make generalizable patterns more easily identifiable and potentially applicable to future projects.

The results of this video observation analysis will be submitted to Professor Raffaele for evaluation.

B. VIDEO OBSERVATION RESEARCH QUESTIONS:

1. **What objects/colors/landmarks attract the most player attention?**
    a. How much time is spent at each object and/or major landmark
        i. Coded* for by: time spent (TS), Success (S / U), questions (QA)
    b. What are players trying to do at each attention center?
2. **How do players react to various in game items/prompts/abilities/obstacles?**
    a. Which ones are clear/unclear for the participant?
        i. Coded* for by: utterance/expression (RVU / RFE), time spent (TS), questions (QA), prompt reaction (PR / PI)
3. **What actions does the player attempt that the game does not allow?**
    a. What confuses the player?
        i. Coded* for by: Utterance(RVU / RFE), Success (S / U), Questions Asked (QA), task accuracy (T / O)

Please find a coding explanation below in section D. Codes and Construct Variables
****Please find a sample coding scheme below in figure 1***

C. Study Design

Video Observation will be used this time in order to try clearly capture participant behavior and retain a record of said behavior for convenient future reference. Rather than standard observation protocols and think-aloud protocols where every behavior can be difficult to record and cannot be accessed again to confirm accuracy, video observation will allow us to keep a detailed record of everything the participant

does in the game. Video observation is also less intrusive and the participant is expected to feel more at ease and relaxed compared to think-aloud protocols.

Conducting video observation does come at somewhat of a cost in that analyzing this recording will take much longer than what might be typically required for standard observation and/or think-aloud analysis. However, we have a crack team of experienced research professionals ready to process this data.

Having a documented and easily re-accessible account of everything researchers detect will make time more time, but ultimately allow for easier future reference. Having detailed and accurate coding from researchers with clear visual observation data from participants will give developers an extremely high resolution look at how participants are interacting with their software.

The participant will be set-up to play through the demo, Europa. They will be given a simple set of tasks and their behaviors as they proceed through said tasks, will be recorded. There will be one recording of gameplay and another recording of the participant's face. The researcher will be using the recording software, OBS Studio.

~~Testing has not yet completed, but the researcher will be using one of the following video recording software options: Camtasia, ScreenFlow, OBS Studio, Datavyu.~~

Participant Personas

~~This participant is a personal relationship. They have little to no experience with video game culture, is 70 years old, identifies as male and his highest level of formal education is college. The participant is also now retired. The participant claims not to play video games on any regular basis and has little to no familiarity with modern game design.~~

~~Given enough time, I would also like to conduct this experiment with another relatively gaming inexperienced participant from a different age demographic to compare and contrast behavior. The target participant would be in their 20s or 30s, identify as female and highest level of education would be university. The participant would likely be working a day job and claim not to play video games on any regular basis nor has any familiarity with modern game design.~~

The participant is in their early 30s, identifies as female and holds a bachelor's degree. She works a day job, has briefly played a video games in the last year, but does not currently play video games on a regular basis, and has experience playing video games in her childhood.

This general information as well as more specific details regarding the participant's background will be captured in varying degrees and recorded in both the pre-test survey and post-test interview. Having access to this extra dimension regarding player background information provides valuable perspective on how specific mechanics, prompts and environmental design choices may affect different players relative to their past gaming experience. The original design did not include a survey or questionnaire, but upon reflection, I think adding those will be a benefit to developers who may want to fine tune the game environment relative to different experience levels.

D. VIDEO OBSERVATION INSTRUMENTS & PROCEDURE

Technical Set-up

1. Prepare the test setup hardware (table, chair, laptop PC with keyboard, trackpad and external wireless mouse, and USB PlayStation 4 controller)
2. Prepare the test setup software (Steam, Europa Demo)
    a. Check controller inputs, video resolution, frame rate, etc. to make sure game is running properly
3. Prepare the video recording hardware (PC, Microphone/Webcam)
4. Prepare the video recording software (~~Camtasia, ScreenFlow,~~ OBS Studio~~, Datavyu~~)
    a. Testing will continue up until session date before selection is made
5. Prepare coding scheme spreadsheet
6. Confirm recording functionality with a 10-second test recording
    a. Make sure both audio and video properly playback
~~a.~~7. Once technical aspects have all been confirmed to be in working order and participant is comfortable, researcher can commence video data collection.

Data Collection Procedural Set-Up

1. Prepare warm up questions to welcome and relax the participant
2. Participant will be reminded about research goals and rights to privacy/confidentiality
3. Participant will be reminded to relax since there is no performance based evaluation
4. Participant will be given pre-test survey
~~3.~~5. Participant will be briefed on think-aloud protocol standards and then given a warm-up word search puzzle activity with which to practice think-aloud verbalization
6. ~~Once technical aspects have been confirmed to be in working order and participant is comfortable, researcher can start recording and commence t~~The main gameplay video recording session will occur.
~~4.~~7. After the gameplay session is complete, the post-test interview will be conducted.

INSTRUMENTS & MEASURES

In order to get a better idea of the participant's background and experience with video games, a brief questionnaire will be administered before the session begins. Additionally, after the main game play session concludes, the participant will be given a post-session exit interview to get more perspective from the participant after they have experienced the game. Neither of these will be video recorded.

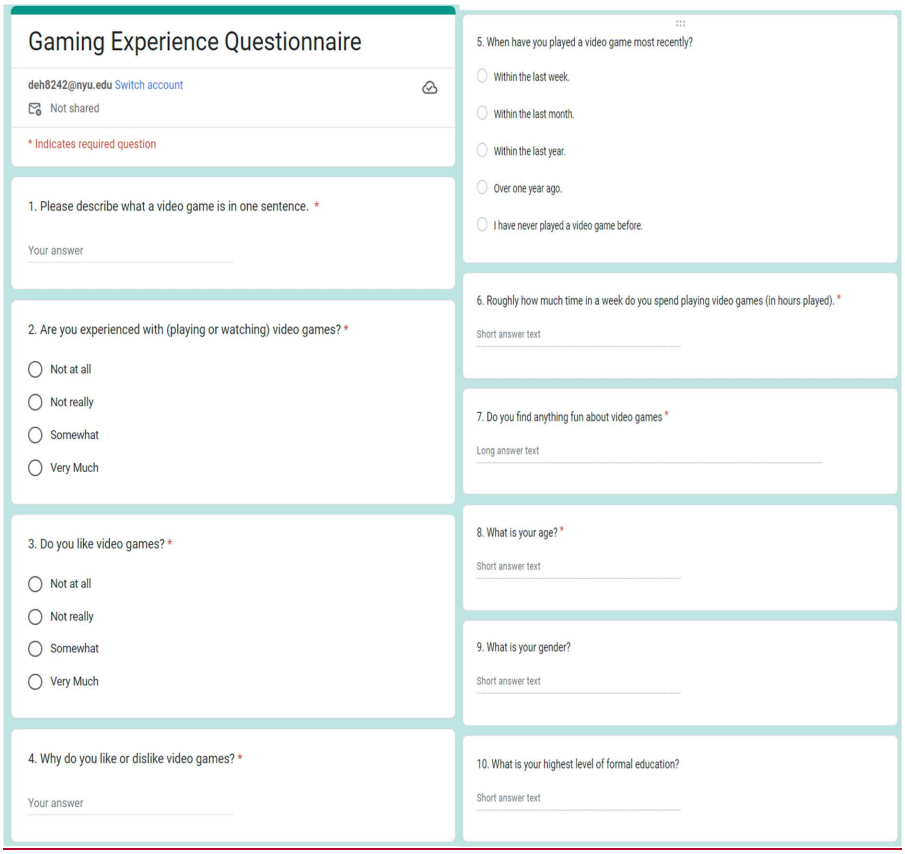
As similarly mention in the persona section, the pre-test survey and post-test interview were not originally a part of my video observation study design. However, after introspection on how to provide more detailed data for researchers and developers it was decided that prior to collecting video data as well as thereafter, it would be beneficial to also collect data that reflects the participants prior knowledge with games in general and their summative reactions to this game experience specifically. Processing this extra data will no doubt be more time-consuming, but having this extra layer will benefit researchers and developers by providing more potential data points from which to draw conclusions and cross reference interesting behavior patterns.

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Gaming Experience Questionnaire

deh8242@nyu.edu Switch account

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* Indicates required question

1. Please describe what a video game is in one sentence. *

Your answer

2. Are you experienced with (playing or watching) video games? *

- Not at all
- Not really
- Somewhat
- Very Much

3. Do you like video games? *

- Not at all
- Not really
- Somewhat
- Very Much

4. Why do you like or dislike video games? *

Your answer

5. When have you played a video game most recently?

- Within the last week.
- Within the last month.
- Within the last year.
- Over one year ago.
- I have never played a video game before.

6. Roughly how much time in a week do you spend playing video games (in hours played). *

Short answer text

7. Do you find anything fun about video games *

Long answer text

8. What is your age? *

Short answer text

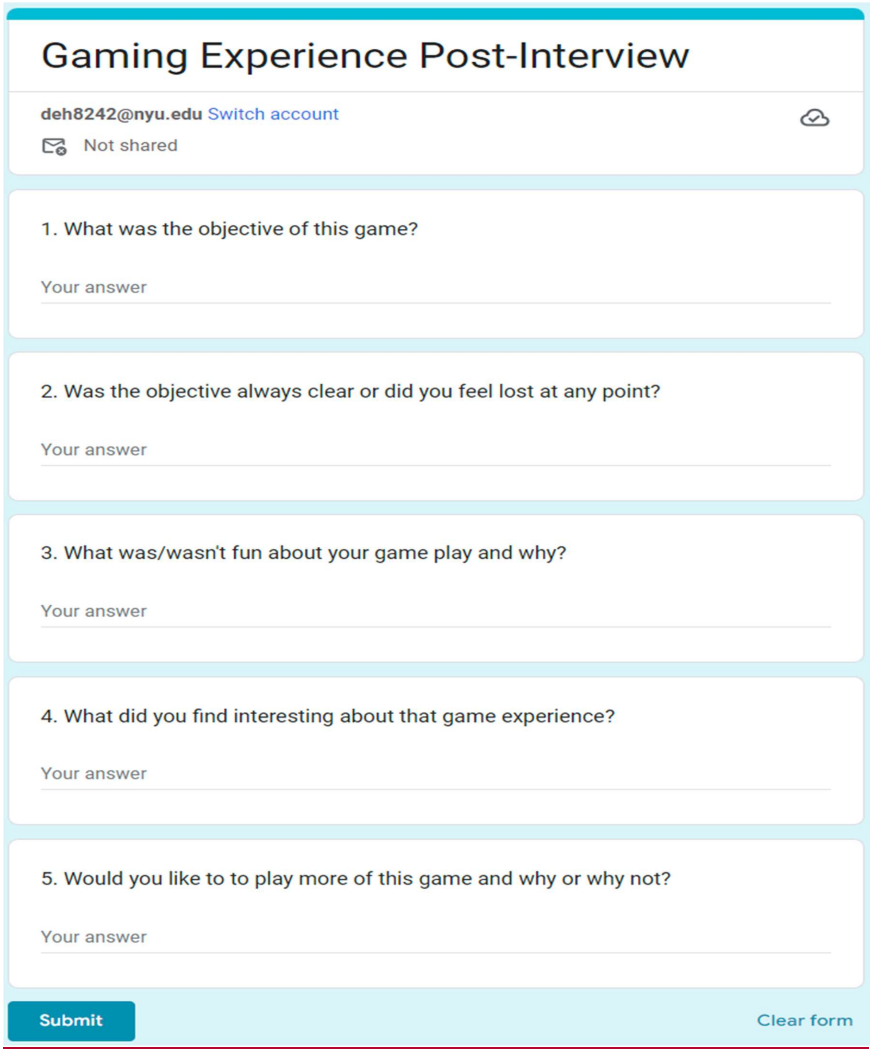
9. What is your gender?

Short answer text

10. What is your highest level of formal education?

Short answer text

sample pre-session questionnaire



sample post-session interview questions

WARM-UP

The participant should be greeted and asked in a welcoming manner to sit down. The main warm up section will begin with a description of what think-aloud protocol is and then a quick demonstration, by the researcher for what think-aloud should look and sound like. Thereafter, an activity will provide the

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participant an opportunity to get comfortable with the think-aloud protocol with a basic word search puzzle. While completing the puzzle, the participant will be appropriately encouraged and reminded to verbalize their thoughts. In order to prime the participant's comfort and ability to think-aloud during the main session, they will be supported with standard think-aloud protocol probes like those listed below:

- What are you thinking now?
- What is happening in this area?
- What do you think about the presentation in this area?
- What is your plan in this area?
- What stands out to you in this section?
- What do you see now?
- What did you just do?
- Why did you do that?
- What are you trying to do next?



sample think-aloud warm up activity

Main Gameplay Session Tasks

~~Because thinking aloud while playing a game isn't necessarily standard behavior for everyone, and the participant may forget to keep talking, some task-specific probes will be provided here for quick reference:~~

Throughout the main gameplay session the participant may be asked to target specific tasks if they stray too far off the critical path for too long. Some of the sample probes to help them get back on track may include:

1. What is the objective here?
2. What are you trying to do now?
3. Where is the goal?

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4. What happens when you (**insert appropriate game action here**)?
~~1.~~5. Do you understand the objective?
~~2.~~6. Please collect the emeralds in this area.
~~3.~~7. Please find the messages in this area.
~~4.~~8. Please obtain the abilities in this area.
~~5.~~9. Please proceed to the next area.

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***Codes and Construct Variables**

- TS - **T**ime **S**pent
  - (duration of **t**ime **s**pent at point of interest) RQ2
- RVU / RFE – **U**tterances / **E**xpressions
  - (**R**eactions **V**erbal **U**tterances/ **R**eactions **F**acial **E**xpression) RQ
- S / U – **S**uccessful / **u**nsuccessful
  - (**S**uccessfully/**u**nsuccessfully understood objective and could/couldn't clear obstacle)
- T / O – On **T**ask or **O**ff Task
  - (Is participant on **t**ask and making progressing towards next objective or **o**ff task meandering)
- QA – **Q**uestions **a**sked
  - (**#** of **q**uestions **a**sked to researcher for help or clarification)
  - (content of **q**uestion **a**sked for assistance)
- PR / PI – In-game **P**rompt **R**ecognized or **P**rompt **I**gnored
  - ~~o~~ When an in-game prompt or message appears, does the participant recognize and react accordingly or ignore the prompt.
  - ○

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As briefly mentioned in the introduction, each construct that was chosen directly correlates to one or more of the research questions of interest. In order to analyze research question 1, what specific items in the environment attract the most player attention, time spent (TS), task success (S/U) and questions asked (QA) will be coded for. To examine research question 2, how players react to specific game elements will be coded for with player reaction (RVU/RFE), prompt reaction (PR/PI), time spent (TS), and questions asked (QA). Regarding question 3, what actions do players attempt that the game or are do not facilitate, will be coded for with task accuracy (T/O), player reaction (RVU/RFE) and questions asked (QA).

Increasing the resolution on the research questions to the point where specific constructs come into view provides a multi-dimensional look at each question. This is not an attempt to quantify qualitative data, but rather an attempt to approach these research questions with more detail. A more granular look at research questions through the lens of these more finely coded constructs allows researchers and developers to potentially draw more points of comparison and/or more colors of contrast. All of which should help developers more clearly identify specific elements that are currently well-designed and elements that may benefit from adjustment.

Sample Coding Scheme Spreadsheet

	A	B	C	D	E	F	G	H	I	J
1	CONSTRUCT		MEASURE		OBSERVABLE BEHAVIOR / INDICATOR		CODE			
2										
3	Time Spent		How much time participant spends @POI		Stopping at a POI		**TS**	**T**ime **S**pent		
4					Stopping anywhere and spending time					
5										
6	TS EXAMPLES	0:45 @ chasing rabbit		1:30 @ruin1		0:20 @tree1				
7		1:25 @ reading book entry		2:00 @bridge1		0:35 @ lantern1				
8										
9	Utterances/Expressions		What does particpant say re: stimulus?		Making remarks		**RVU / RFE**	**R**eaction **V**erbal **U**tterance /		
10			What expression does participant make?		Making facial expressions			**R**eaction **F**acial **E**xpression		
11										
12	RVU EXAMPLES	"This reminds me of Zelda"		"ooh, look at that"		"Oh, wait I collected something"				
13	RVE EXAMPLES	Smile		Eyes Widen		Eyes narrow				
14										
15	Successful/Unsuccessful Task		Can participant complete tasks?		Can partipant explain objective?		**S / U**	**S**uccessful/**U**nsuccessful		
16					How many emeralds found?					
17										
18	S EXAMPLES	Found emerald		lit lantern		cleared first area				
19	U EXAMPLES	found all emeralds		used power drop to light multiple lanterns						
20										
21	On or Off Task		Is participant proceeding to task?		Is participants proceeding towards next POI?		**T / O**	On-**T**ask / **O**ff-Task		
22			Or is partiants meandering away from task		Is participant moving away from next POI?					
23										
24	T EXAMPLES	found and followed each beam of light correctly			was able to light the lanterns and open area doors					
25	O EXAMPLES	found one emerald but didn't continue looking			did not follow prompt to execute stomp maneuver					
26										
27	Questions Asked		What questions does the participant want		# of questions participant asks		**QA**	**Q**uestions **A**sked		
28					Theme or pattern of questions asked					
29										
30	QA EXAMPLES	"How do you get over there?"			"Am I from Earth?"					
31										
32	Prompt Reactions		How does participant respond to game pr		Does participant do anything in reaction to prompt?		**PR / PI**	**P**rompt **R**ecognized /		
33					What does participant say/do after reading prompt?			**P**rompt **I**gnored		
34	PI EXAMPLES	"Press X to interact"								
35	PR EXAMPLES	"Press A to jump" ---reaction--- Glanced down at the controller				"stomping creates larger bubble" -- stopped moving and squinted				

Figure 1, sample coding scheme

I already had a sample coding scheme in the original design, but I did a brief review of the video data for this reflection and while filling it out for practice, I realized that the form could use some adjustments. Though I think all of these constructs/measures do help to get closer to answering the research questions, the way the form is currently constructed is slightly constricting to use in practice. During the session I had the spreadsheet printed out on one landscape formatted page, but there isn't enough space for more than a few remarks. In future versions, I would either print out the form in parts, separated by construct for easier transcription by hand or I would try to only fill in the form digitally within spreadsheet program like excel so I could easily add space as needed.

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	A	B	C	D	E	F	G	H	I	J
1	CONSTRUCT		MEASURE		OBSERVABLE BEHAVIOR / INDICATOR		CODE			
2										
3	Time Spent		How much time participant spends @POI		Stopping at a POI		**TS**	**T**ime **S**pent		
4					Stopping anywhere and spending time					
5										
6	Utterances/Expressions		What does particpant say re: stimulus?		Making remarks		**RVU / RFE**	**R**eaction **V**erbal **U**tterance /		
7			What expression does participant make?		Making facial expressions			**R**eaction **F**acial **E**xpression		
8										
9	Successful/Unsuccessful Task		Can participant complete tasks?		Can partipant explain objective?		**S / U**	**S**uccessful/**U**nsuccessful		
10					How many emeralds found?					
11										
12	On or Off Task		Is participant proceeding to task?		Is participants proceeding towards next POI?		**T / O**	On-**T**ask / **O**ff-Task		
13			Or is partiants meandering away from task?		Is participant moving away from next POI?					
14										
15	Questions Asked		What questions does the participant want answered?		# of questions participant asks		**QA**	**Q**uestions **A**sked		
16					Theme or pattern of questions asked					
17										
18	Prompt Reactions		How does participant respond to game prompts?		Does participant do anything in reaction to prompt?		**PR / PI**	**P**rompt **R**ecognized /		
19					What does participant say/do after reading prompt?			**P**rompt **I**gnored		

~~***Figure 1, sample coding scheme***~~

E. DATA ANALYSIS PLAN

After the session is complete the video data will be saved and backed up to another drive for emergency backup. A minimum crack team of at least two researchers will then simultaneously commence analysis by reviewing video data while coding for target constructs. Upon completion of video review and construct coding, each researcher will cross-reference their results to solidify researcher inter-reliability. Researchers A and B can also compare notes to expose any inconsistencies and to confirm any potential patterns of interest and/or correlations in observed participant behavior.

These results will be then be written up into a brief report that includes the participant's profile, the coding scheme used, the instruments used, the observation procedure employed as well as any utterances, reactions or other video excerpts that demonstrate any particularly interesting correlations. Upon request, a brief slide show presentation consolidating all of this data can additionally be prepared and delivered.

REFLECTION CONCLUSION

This video observation document was designed to help analyze how well the upcoming game, Europa communicates it's objectives to players. In order to operationalize how well the game communicates to players and how well players comprehend the game design, three specific research questions were posed: 1) **What objects/colors/landmarks attract the most player attention?** (Question 1 helps identify what objects stand out most to players which implies what can be used most reliably for leading the player through the environment. **2) How do players react to various in game items/prompts/abilities/obstacles?** (Question 2 helps show what responses to certain in game-events prompt what player actions so that congruence between game element and player behavior can be confirmed. And **3) What actions does the player attempt that the game does not allow?** (Question 3 helps expose what potential miscommunication may occur between the game and the player.

In order to further break down these complex questions into digestible components, a 6 part coding scheme was devised and implemented within this study design to quantify the answers to the research questions. The constructs of focus include, time spent, participant reaction, task success, task accuracy, questions asked, and prompt recognition. With each research question directly correlating to a number of constructs, the coding scheme, in turn, relates to player behavior and comprehension. The results of this analysis will provide researchers and developers with high resolution data analysis regarding how well their game's environment and its mechanics communicate to the player. This analysis also provides insights into potential miscommunications that may occur between the player and the game, as well as possible assumptions or misunderstandings the player may come into the game experience with. Even if such preconceptions may not have been inspired by this game, useful insights might still be provided for possible future adjustments that could benefit the game.

Overall, I think this study design helps approach answers to questions regarding how clearly the game, Europa, communicates game objectives because the constructs that I decided to code for all represent specific ways to look at player attention and comprehension. Zooming out a bit, I did overlook the potential benefits of surveys and think-aloud in my original design, so I made sure to include them in my revision. Regarding practicality, my original coding sheet was okay, but I've noted the utility of rearranging the format by construct for transcribing by-hand or to simply fill it out digitally. It is certainly more time-consuming, but I think a combination of all three methods we've practiced thus far, would yield the richest participant data to get closest to answering the research questions.

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